

Foundations of R Chapter 10

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```
library(dplyr, warn.conflicts = F)
library(stringr)
library(ggplot2)
library(readr)
library(qdapDictionaries)
library(purrr, warn.conflicts = F)
```

This chapter introduces a study the wage gap between men and women among Chicago's public employees and learn a bit about matrix notation in R.

The following code reads in data from Open Chicago, a data source on Chicago city information.

```
employees <- read_csv("~/Dropbox/mathCamp/Code/data/CurrentEmployees.csv", col_types = "cccc")
employees <- employees %>% mutate(salarynum = as.numeric(str_extract(`Employee Annual Salary`,
  "[0-9]+")), nolastname = str_extract(Name, "(?<=,).+"), firstname = str_extract(nolastname,
  "[^\\s]+"))
```

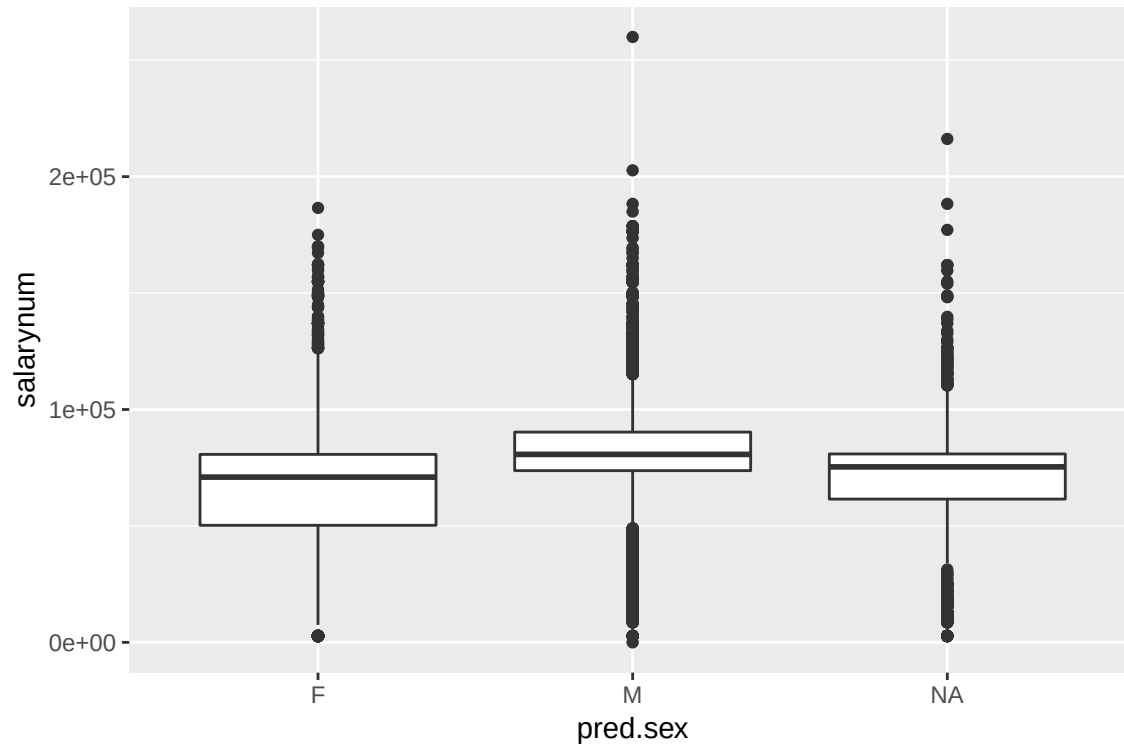
The qdapDictionaries library has a file called NAMES, which include columns including name and pred.sex, which is the predicted binary gender of each name. Here we join by exact matches.

```
UNAMES <- unique(NAMES[, c("name", "pred.sex", "gender2")])
names(UNAMES) <- c("firstname", "pred.sex", "gender2")
codedgender <- employees %>% left_join(UNAMES)
```

```
## Joining, by = "firstname"
```

From here we can begin to study the differences in salary across genders.

```
ggplot(codedgender, aes(pred.sex, salarynum)) + geom_boxplot()
```



```
justcoded <- codedgender %>% select(pred.sex, salarynum) %>% na.omit()

justcoded %>% group_by(pred.sex) %>% summarise(meansalary = mean(salarynum), sdsalary = sd(salarynum))

## # A tibble: 2 x 3
##   pred.sex meansalary sdsalary
##   <fct>      <dbl>    <dbl>
## 1 F          64993.    27804.
## 2 M          80838.    19130.

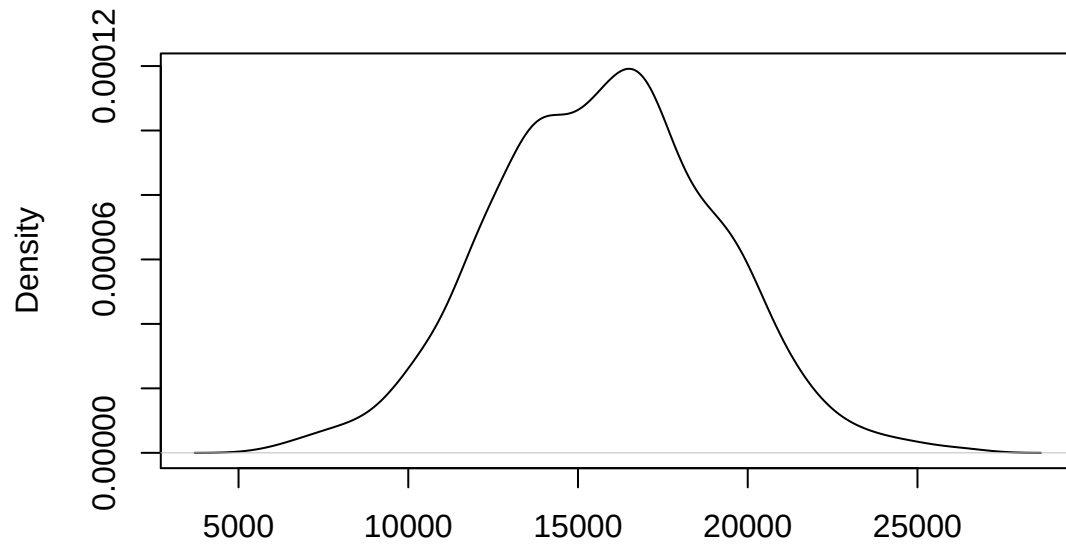
meansamples <- c()
for (i in 1:1000) {

  mensalarys100 <- sample(justcoded$salarynum[justcoded$pred.sex == "M"], 100)
  womensalarys100 <- sample(justcoded$salarynum[justcoded$pred.sex == "F"], 100)

  meansamples[i] <- mean(mensalarys100) - mean(womensalarys100)
}

plot(density(meansamples), main = "100 person samples")
```

100 person samples



N = 1000 Bandwidth = 744.8

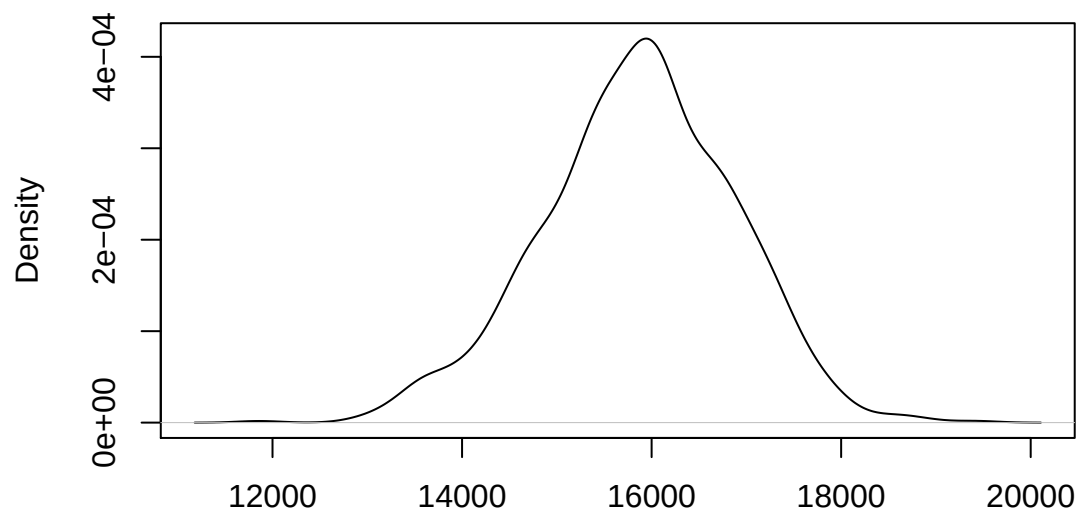
```
meansamples1k <- c()
for (i in 1:1000) {

  mensalarys1k <- sample(justcoded$salarynum[justcoded$pred.sex == "M"], 1000)
  womensalarys1k <- sample(justcoded$salarynum[justcoded$pred.sex == "F"], 1000)

  meansamples1k[i] <- mean(mensalarys1k) - mean(womensalarys1k)
}

plot(density(meansamples1k), main = "1k person samples")
```

1k person samples



N = 1000 Bandwidth = 226.7

We can do similar analysis Using (perhaps too much) dplyr

```
ttestout<-justcoded %>%
```

```
  summarise(

    gapestimate = diff(t.test(salarynum[pred.sex == "M"],
                              salarynum[pred.sex == "F"],
                              var.equal = T )$estimate),

    tstat = t.test(salarynum[pred.sex == "M"],
                   salarynum[pred.sex == "F"],
                   var.equal = T )$statistic,

    df = t.test(salarynum[pred.sex == "M"],
                 salarynum[pred.sex == "F"],
                 var.equal = T )$parameter,

    pvalue=1-pt(tstat, df))
```

```
ttestout
```

```
## # A tibble: 1 x 4
##   gapestimate tstat    df pvalue
##         <dbl> <dbl> <dbl>   <dbl>
## 1    -15845.  55.9 29098     0
```

```
summary(lm(salarynum ~ pred.sex, data = justcoded))
```

```
##
## Call:
## lm(formula = salarynum ~ pred.sex, data = justcoded)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -80838  -9090   -114   12354  179166
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  64993.3      238.4   272.57  <2e-16 ***
## pred.sexM    15844.6      283.7    55.86  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22030 on 29098 degrees of freedom
## Multiple R-squared:  0.09684,    Adjusted R-squared:  0.09681
## F-statistic:  3120 on 1 and 29098 DF,  p-value: < 2.2e-16
```

It seems that women earn 15,800 dollars less, on average, than men.

Now we might be worried that we have unclassified men and women. Suppose we had a simple model for names, where $p(\theta_i|\theta_{i-1})$ is the probability of a letter i following letter $i - 1$. We can then populate a matrix of letter to letter transitions. For example, is a “B” likely to follow a “Q”? Probably not. I add two spaces for the first and last letter respectively. We can then estimate these probabilities from the labeled names.

```
Female <- UNAMES %>% filter(pred.sex == "F")
lettersinnamesF <- str_split(Female$firstname, "")

lettersmat <- matrix(rep(0, 28 * 28), nrow = 28)

pupdate <- function(lnames) {
  letternums <- c(27, unlist(map(lnames, function(x) {
    which(LETTERS == x)
  })))
  letternumfollow <- c(letternums[-1], 28)
  lettersmat[letternums, letternumfollow] <- 1
  return(lettersmat)
}

mappingsF <- map(lettersinnamesF, pupdate)

finalmatF <- Reduce("+", mappingsF)

letterrowsumF <- rowSums(finalmatF)

finalnormF <- finalmatF/letterrowsumF
```

Here we have the full transition matrix for female names.

We can repeat the same process for male names.

```
Male <- UNAMES %>% filter(pred.sex == "M")
lettersinnamesM <- str_split(Male$firstname, "")

lettersmat <- matrix(rep(0, 28 * 28), nrow = 28)

pupdate <- function(lnames) {
  letternums <- c(27, unlist(map(lnames, function(x) {
    which(LETTERS == x)
  })))
  letternumfollow <- c(letternums[-1], 28)
  lettersmat[letternums, letternumfollow] <- 1
  return(lettersmat)
}

mappingsM <- map(lettersinnamesM, pupdate)

finalmatM <- Reduce("+", mappingsM)

letterrowsumM <- rowSums(finalmatM)
finalnormM <- finalmatM/letterrowsumM
```

Ok, so for the two genders, the probability of each letter following a B is:

```
finalnormM[2, ]
```

```
## [1] 0.060723514 0.158914729 0.016795866 0.025839793 0.089147287
## [6] 0.005167959 0.005167959 0.009043928 0.043927649 0.002583979
## [11] 0.009043928 0.040051680 0.010335917 0.054263566 0.062015504
## [16] 0.001291990 0.000000000 0.109819121 0.015503876 0.056847545
## [21] 0.031007752 0.000000000 0.005167959 0.000000000 0.028423773
## [26] 0.000000000 0.000000000 0.158914729
```

```
finalnormF[2, ]
```

```
## [1] 0.0909614277 0.1560161197 0.0103626943 0.0213010938 0.1105354059
## [6] 0.0011514105 0.0092112838 0.0213010938 0.0759930915 0.0000000000
## [11] 0.0120898100 0.0529648820 0.0161197467 0.0500863558 0.0230282096
## [16] 0.0017271157 0.0000000000 0.0817501439 0.0178468624 0.0408750720
## [21] 0.0115141048 0.0028785262 0.0005757052 0.0000000000 0.0310880829
## [26] 0.0046056419 0.0000000000 0.1560161197
```

Awesome huh? Notice that there is 0 probability of the character reserved for the beginning of a word following a B. Checking our intuition from earlier, the probability that a “B” follows a “Q” is rightfully 0. Now we can see if this works. Suppose we test the accuracy.

```
namestest <- c(sample(Male$firstname, 500), sample(Female$firstname, 500))
```

```
namecut <- str_split(namestest, "")
```

```
pcheck <- function(lnames) {
  letternums <- c(27, unlist(map(lnames, function(x) {
    which(LETTERS == x)
  })))
  letternumfollow <- c(letternums[-1], 28)
  fprob <- prod(diag(finalnormF[letternums, letternumfollow]))
  mprob <- prod(diag(finalnormM[letternums, letternumfollow]))
  fprob <- fprob/(fprob + mprob)
  return(fprob)
}
```

```
probs <- unlist(map(namecut, pcheck))
```

```
table(c(rep("M", 500), rep("F", 500)), ifelse(probs > 0.5, "Flab", "Mlab"))
```

```
##
##      Flab Mlab
## F   367  133
## M   166  334
```

Seems that we don't always get it wrong! My own name (BOBBY) occurs as a male name with probability 0.9044971. ROBERT is male with probability 0.8309851.

The most male name in the dataset is MAXWELL, the least male name (of male names) is NATHANIAL.

```
Male$probF <- unlist(map(str_split(Male$firstname, ""), pcheck))
Male %>% arrange(probF) %>% head(3)
```

##	firstname	pred.sex	gender2	probF
## 1	MAXWELL	M	M	0.000000000
## 2	WOODROW	M	M	0.008434792
## 3	WILFREDO	M	M	0.019605808

```
Male %>% arrange(probF) %>% tail(3)
```

##	firstname	pred.sex	gender2	probF
## 1144	ISAIAH	M	M	0.9022252
## 1145	ISAIAS	M	M	0.9027460
## 1146	NATHANIAL	M	M	0.9102413

Among the most Female name is JACKQUELINE, the least Female name is WILLOW, followed by BUFFY (coincidence?).

```
Female$probF <- unlist(map(str_split(Female$firstname, ""), pcheck))
Female %>% arrange(probF) %>% tail(3)
```

##	firstname	pred.sex	gender2	probF
## 4014	ROXY	F	F	1
## 4015	XUAN	F	F	1
## 4016	JACKQUELINE	F	F	1

```
Female %>% arrange(probF) %>% head(3)
```

##	firstname	pred.sex	gender2	probF
## 1	WILLOW	F	F	0.03935616
## 2	BUFFY	F	F	0.05582483
## 3	MARJORY	F	F	0.07290241

We can use a more conservative coding to try to rule out making guesses.

```
namestest <- c(sample(Male$firstname, 500), sample(Female$firstname, 500))

namecut <- str_split(namestest, "")
probs <- unlist(map(namecut, pcheck))

table(c(rep("M", 500), rep("F", 500)), ifelse(probs > 0.6, "Flab", ifelse(probs <
  0.4, "Mlab", "Nolab")))
```

```
##
##      Flab Mlab Nolab
##  F  271   75   154
##  M   86  233   181
```

Great, this way we get substantially lower errors, at the cost of not coding about a third of the data. This indicates a tradeoff - we can either make more mistakes, or we can leave out data.

Using this formulation, we can try to classify the uncoded genders as follows:

```
uncoded <- codedgender %>% filter(!pred.sex %in% c("F", "M"))
namecut <- str_split(uncoded$firstname, "")
uncoded$probsF <- unlist(map(namecut, pcheck))
```

```
uncoded %>% select(Name, firstname, probsF) %>% arrange(probsF) %>% head()
```

```
## # A tibble: 6 x 3
##   Name                firstname probsF
##   <chr>              <chr>      <dbl>
## 1 DEL MUNDO, ROCKWELL R ROCKWELL  0
## 2 HO, KWOK L        KWOK      0
## 3 MILLAN, BJORN M    BJORN     0
## 4 NIZAMUDDIN, SHOWKET F SHOWKET  0
## 5 LITTLE, DURWARD J  DURWARD  0.0213
## 6 RODRIGUEZ, SIGFREDO SIGFREDO  0.0240
```

```
uncoded %>% select(Name, firstname, probsF) %>% arrange(desc(probsF)) %>% head()
```

```
## # A tibble: 6 x 3
##   Name                firstname probsF
##   <chr>              <chr>      <dbl>
## 1 BALATA, TOMASZ      TOMASZ      1
## 2 BANACHOWSKI, FRANCISZEK FRANCISZEK  1
## 3 CHLEBOWICZ, LESZEK  LESZEK      1
## 4 CHOJNACKI, MARIUSZ A MARIUSZ      1
## 5 CUNNINGHAM, RAQHEL  RAQHEL      1
## 6 CWIAKALA, TOMASZ    TOMASZ      1
```

```
summary(uncoded$probsF)
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
## 0.0000 0.3721 0.5569 0.5363 0.7015 1.0000     4
```

Finally, we can say, for those that are more than 60 percent likely to be male, call them male, for those that are more than 60 percent likely to be female, call them female.

```
namecut <- str_split(codedgender$firstname, "")
codedgender$probsF <- unlist(map(namecut, pcheck))
codedgender <- codedgender %>% mutate(newGender2 = ifelse(pred.sex %in% c("M", "F"),
  as.character(pred.sex), ifelse(probsF > 0.6, "F", ifelse(probsF < 0.4, "M", NA))))
table(codedgender$newGender2, codedgender$pred.sex, useNA = "always")
```

```
##
##           F      M <NA>
## F      8537      0 1311
## M         0 20563   886
## <NA>      0       0  863
```

```
summary(lm(salarynum ~ newGender2 == "F", data = codedgender))
```

```
##
## Call:
## lm(formula = salarynum ~ newGender2 == "F", data = codedgender)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -80582  -9494    142   12610  179422
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      80581.6      151.9   530.50 <2e-16 ***
## newGender2 == "F"TRUE -15558.2      270.8   -57.46 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22250 on 31295 degrees of freedom
## (863 observations deleted due to missingness)
## Multiple R-squared:  0.09542,    Adjusted R-squared:  0.09539
## F-statistic: 3301 on 1 and 31295 DF,  p-value: < 2.2e-16
```

OK, so given all of this, can we simulate a male and female name? YES!

Lets find two names that start with the letter M.

```
simMname <- "M"

letterdraw <- "M"
for (i in 1:10) {
  if (letterdraw == ".") {
    break
  } else {
    letterdraw <- sample(c(LETTERS, "."), prob = finalnormM[which(LETTERS ==
      letterdraw), c(1:26, 28)], size = 1)
    simMname <- paste(simMname, letterdraw, sep = "")
  }
}
simMname
```

```
## [1] "MEDOU."
```

```
simFname <- "M"

letterdraw <- "M"
for (i in 1:10) {
  if (letterdraw == ".") {
    break
  } else {
    letterdraw <- sample(c(LETTERS, "."), prob = finalnormF[which(LETTERS ==
      letterdraw), c(1:26, 28)], size = 1)
    simFname <- paste(simFname, letterdraw, sep = "")
  }
}
simFname
```

```
## [1] "MASIL."
```

I would not name my kid MEDOU. or MASIL., but if you did, it would be a very masculine/feminine name (according to our simple model).

Problems

```
URLchunkA <- "https://raw.githubusercontent.com/washingtonpost/"
URLchunkB <- "data-police-shootings/master/fatal-police-shootings-data.csv"
shootings <- read_csv(paste0(URLchunkA, URLchunkB))

## Parsed with column specification:
## cols(
##   id = col_integer(),
##   name = col_character(),
##   date = col_date(format = ""),
##   manner_of_death = col_character(),
##   armed = col_character(),
##   age = col_integer(),
##   gender = col_character(),
##   race = col_character(),
```

```
## city = col_character(),
## state = col_character(),
## signs_of_mental_illness = col_character(),
## threat_level = col_character(),
## flee = col_character(),
## body_camera = col_character()
## )
```

- 1) This question asks us to use ggplot to display the timing of police shootings data above.
 - a) First, use the **lubridate** package to round dates to the nearest month. Use the **round_date** function.
 - b) Display these data with a histogram with 10 bins, 20 bins, 30 and 35 bins. Do the same with a density plot with varying adjustment equal to .1, .2, and .8. Describe what is happening.
 - c) Produce a table which displays the count of white people that were shot and the presence of a body camera.
 - d) What would the expected counts be if these two variables (being white and body camera presence) were independent?
 - e) What is the difference between the observed counts and the expected counts?
 - f) Square the answers from e), divide by the expected count, and take the sum. This is called the chi-square statistical test of independence. It is distributed chi-square with $2*2 - 1 = 3$ degrees of freedom.
 - g) Use the cdf function for the chi-square to describe the probability of observing the result from f) were these two variables independent.
 - h) Replicate the above steps with the **chisq.test** command.
- 2) Use the **lakers** dataset from the lubridate package to determine the following:
 - a) Which top 40 Laker's performance (in terms of made free throw shots) was most affected by being home or away?
 - b) Which top 40 Laker's performance improved most (in terms of made free throw shots) when the lakers were losing?
 - c) **ADVANCED/Extra Credit:** Plot the distribution of time between shots over the course of games.